Logistics & Supply Chain Analytics

Data cleaning, KPI & Performance Analysis Report

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Note on data: This report is built directly from the record-level data stored inside the Power BI file (n.pbix) — 5,767 cross-border orders placed between 1 January 2023 and 29 August 2023, covering 5 product categories, 5 origin and 5 destination countries, 3 shipping modes (Air, Sea, Land), 3 customer segments and 49 warehouses. The dataset contains no missing values or duplicate Order IDs. Every KPI, chart value and crosstab below was computed from those rows, and the charts are re-drawn from the same data as the dashboard visuals (the dashboard itself is built from 4 pages: Executive, Logistics, Inventory and Dataset).

KEY WORDS:-

- Revenue & Product Mix
- Transport & Shipping Mode
- Customs & Geography
- Delay & Time Slot
- Warehouse Capacity
- Stock Risk & Reorder
- Demand & Promotion

Objective

This report answers seven recurring questions that a supply chain and logistics control team would ask of a cross-border operations dashboard, using descriptive analytics only — it summarizes what the data shows rather than forecasting what happens next.

1. How is order value distributed, and do certain product categories or customer segments generate disproportionately more revenue?

2. How do Air, Sea and Land shipping compare, and does the choice of transport mode change delivery lead time or on-time performance?

3. Where do shipments go, and do customs clearance times create bottlenecks for particular destination countries?

4. Does shipping distance explain delivery lead time, or is delivery speed driven by something other than how far goods travel?

5. How often are orders delayed, and do delays concentrate in particular time slots or months?

6. How full are the warehouses, and which product categories carry the most stock risk (stock at or below the reorder point)?

7. How do demand, seasonality and promotions interact with these operational measures when compared against each other rather than viewed individually?

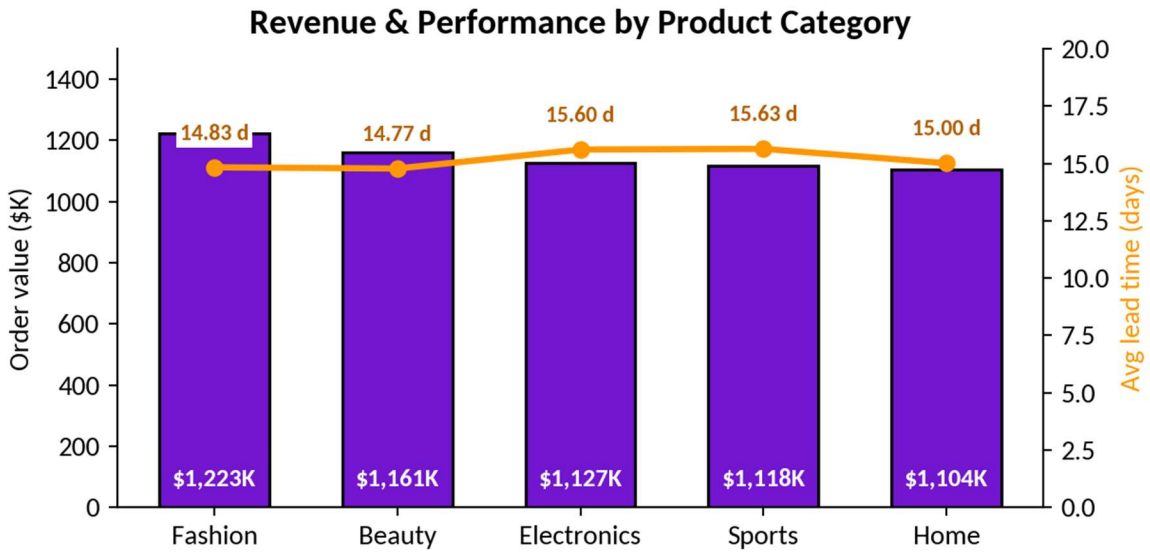
How This Would Work at Real Scale

Tracking orders, carrier events, customs status and warehouse stock for a live global network is an engineering problem before it is an analytics one. A realistic architecture for a production cross-border supply chain analytics pipeline:

Layer	Purpose	Typical Tools
Ingestion Layer	Capture orders, carrier tracking events, customs filings and warehouse stock movements as they happen	ERP / WMS / TMS exports, carrier and customs APIs
Stream Processing	Roll raw shipment events up into per-lane, per-warehouse and per-day aggregates (lead time, delay flags, utilization)	Stream Processing (Kafka / Spark-style jobs)
Storage Layer	Store cleaned order-level records (order, shipment, inventory, promotion fields) for portability and downstream analysis	Data Platform (CSV / warehouse tables)
Serving / BI Layer	Dashboards with slicers for category, segment, shipping mode and month, refreshed hourly or daily for executive, logistics and inventory teams	Power BI

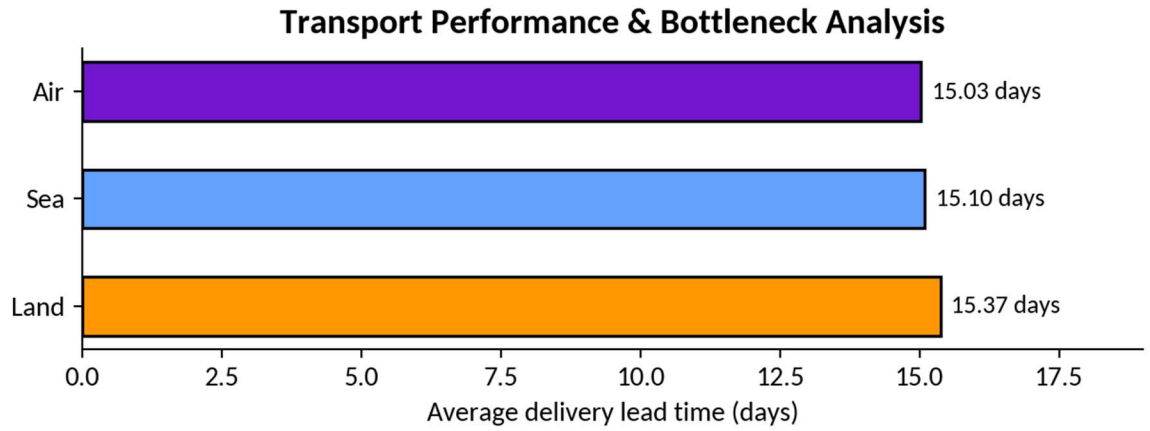
The report above reflects the output of that pipeline — an order-level dataset of 5,767 records — not raw carrier telemetry, since no dashboard renders millions of individual tracking pings usefully. The same chart structure would apply directly once a live shipment-event stream is substituted in.

Revenue & Performance by Product Category



Order value is spread almost evenly across the five product categories rather than concentrated in a single line. Fashion generates the highest value with $1.22M (21.3% of the $5.73M total) across 1,212 orders, followed by Beauty at $1.16M, Electronics at $1.13M, Sports at $1.12M and Home at $1.10M. The gap between the largest (Fashion) and smallest (Home) category is $118.7K, only 2.1 percentage points of total value. The performance line is just as flat: average delivery lead time runs from 14.77 days (Beauty) to 15.63 days (Sports), a spread of 0.86 days. The five category totals sum to $5,732,623, which reconciles with the Total Order Value KPI. For planning, this means no single category is a revenue-concentration risk, and logistics service levels do not need to be designed around one product line.

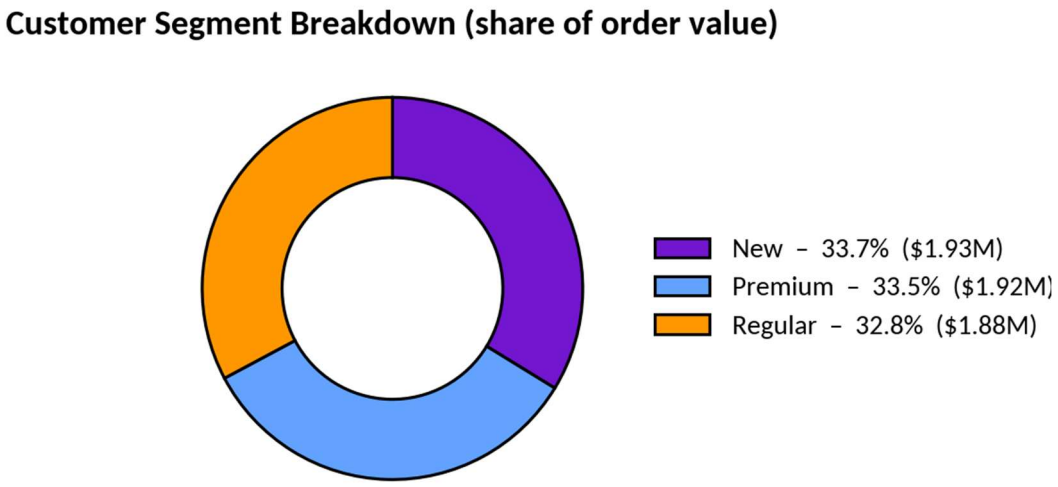
Transport Performance & Bottleneck Analysis



Average delivery lead time is almost identical across the three shipping modes. Land is the slowest at 15.37 days, followed by Sea at 15.10 days and Air at 15.03 days — a gap of just 0.35 days (about 8 hours) between the fastest and slowest mode. This is a notable finding in itself: Air freight, normally expected to

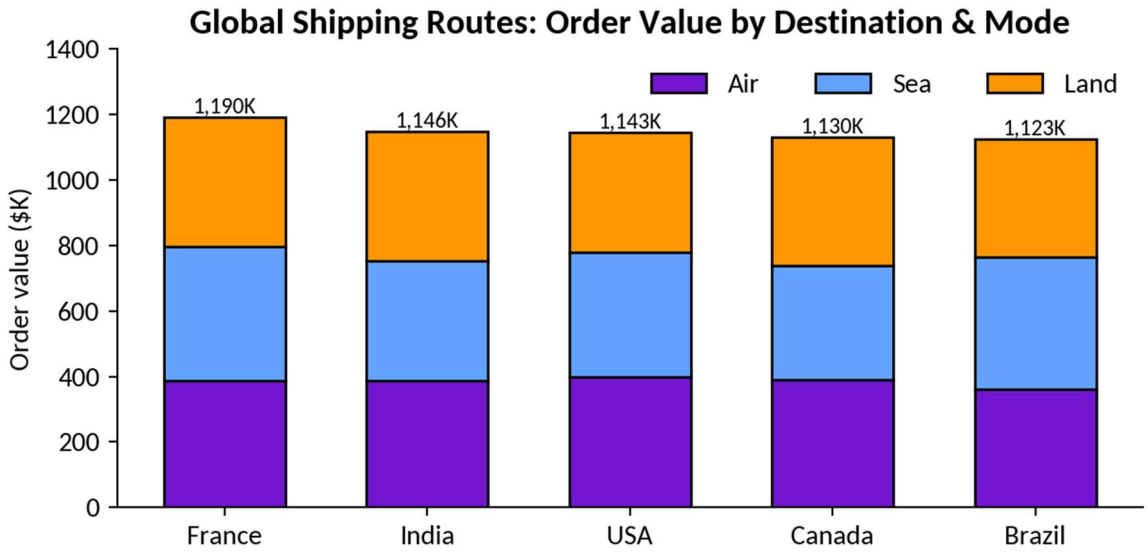
be far quicker than Sea, delivers only marginally faster in this dataset, so the choice of transport mode is not what drives lead time here. Order volume is also balanced — Air handles 1,953 orders, Sea 1,942 and Land 1,872 — and the overall average lead time is 15.16 days. The operational implication is that bottlenecks are more likely to sit in processes common to all modes (customs, handling, warehouse dispatch) than in the transport leg itself.

Customer Segment Breakdown



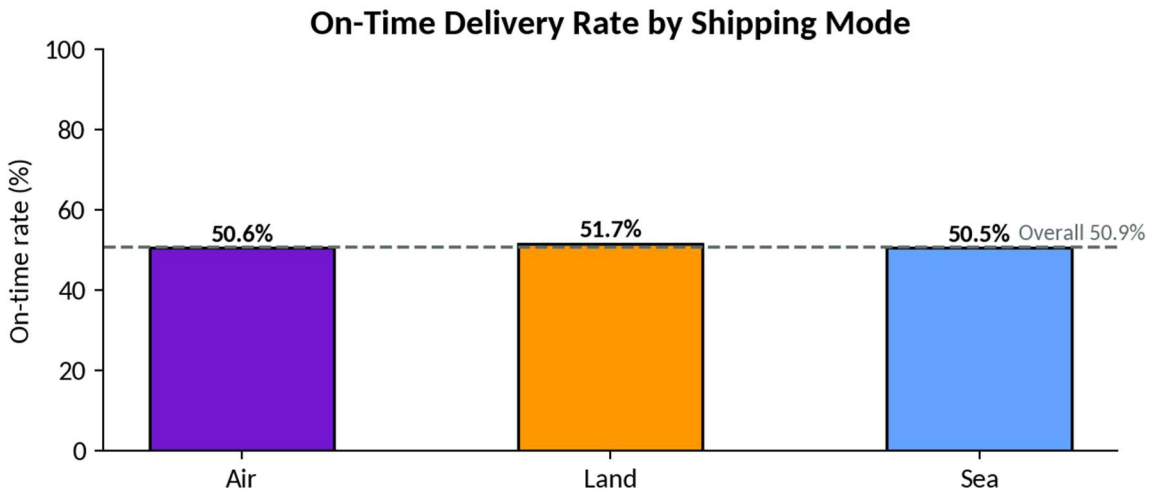
Order value is split almost evenly across the three customer segments. New customers contribute $1.93M (33.7%), Premium $1.92M (33.5%) and Regular $1.88M (32.8%). Order counts follow the same pattern (1,954 New, 1,929 Premium and 1,884 Regular), and average order value is close to $1,000 in every segment (990 to 998 dollars). Premium customers do not spend more per order than New or Regular customers in this sample, which suggests the Premium label reflects service level rather than basket size. The only visible difference is delay exposure: 50.6% of Premium orders were delayed versus 47.3% of Regular orders, so the segment expected to receive the best service experiences the most delays.

Global Shipping Route Map



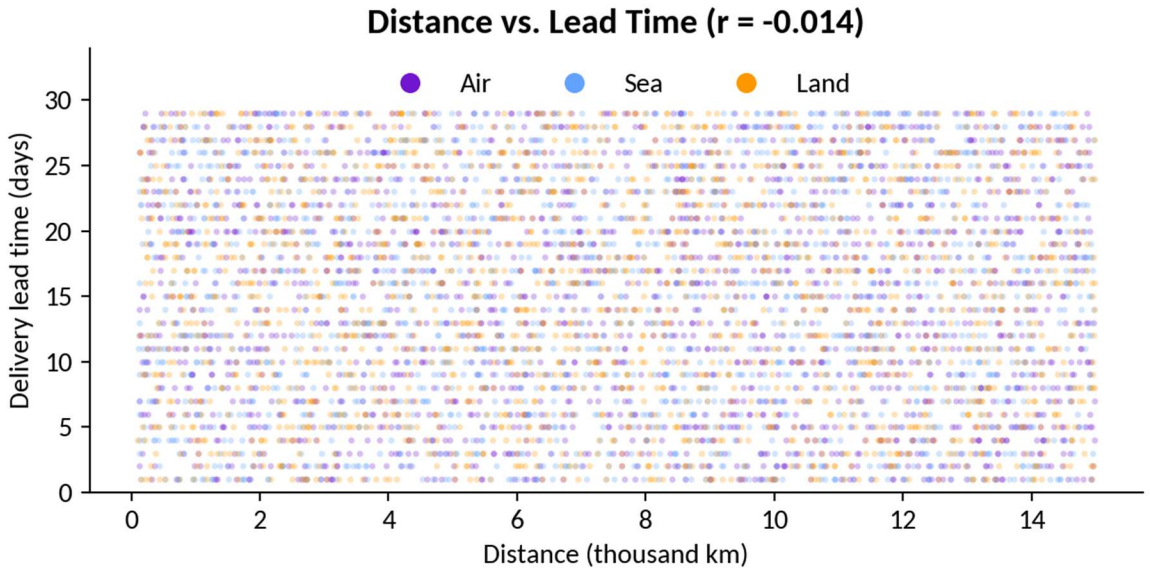
Order value is distributed almost uniformly across the five destination countries, and each destination is served by all three shipping modes in similar proportions. France is the largest destination with $1.19M, followed by India ($1.15M), USA ($1.14M), Canada ($1.13M) and Brazil ($1.12M). The gap between the largest and smallest destination is $66.4K. The origin side is equally balanced (China, Germany, India, UK and USA each originate between 1,135 and 1,181 orders), and 442 orders (7.7%) are shipped to the same country they originate from. Because the dashboard's map cannot be reproduced in a static report, this chart shows the same fields (destination, shipping mode, order value) as stacked bars. Route planning should therefore treat all five destination markets as comparable in volume.

On-Time Delivery Rate by Shipping Mode



Only about half of all orders arrive on time. The overall on-time rate is 50.9%, which means 2,831 of 5,767 orders (49.09%) were delayed. By mode, Land performs best at 51.7%, followed by Air at 50.6% and Sea at 50.5%, a spread of only 1.1 percentage points. No transport mode is meaningfully more reliable than another, and none comes close to a typical service-level target such as 90–95% on-time delivery. Delay is therefore a network-wide issue rather than a problem with a specific mode. Corrective effort should concentrate on the factors shared by every shipment — handling, customs processing and dispatch scheduling — rather than on switching carriers or modes.

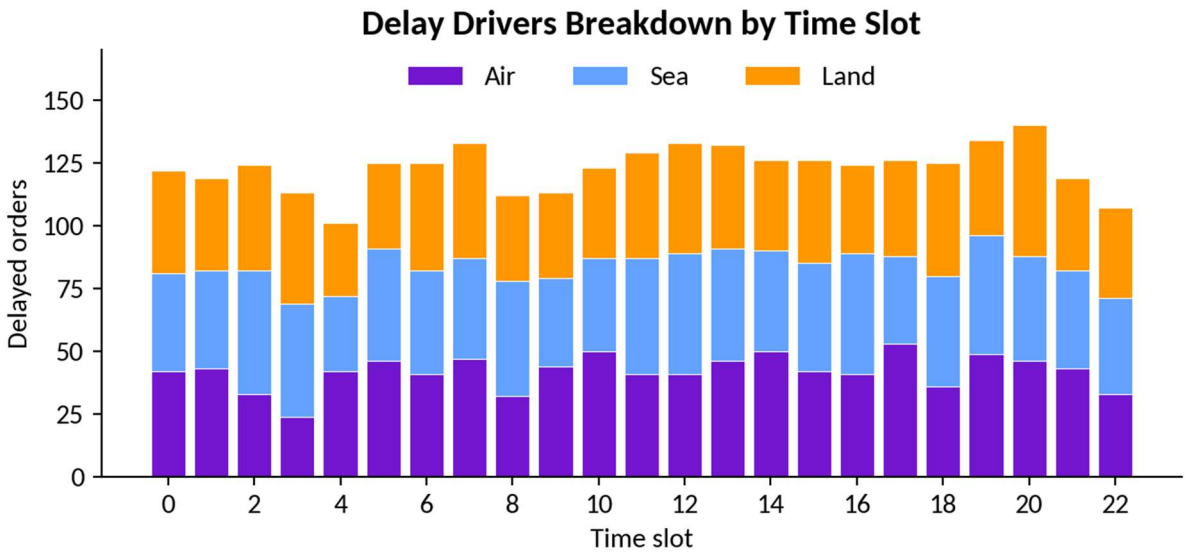
Distance vs. Lead Time



Shipping distance has essentially no relationship with delivery lead time in this dataset. Across all 5,767 orders the correlation between distance and lead time is -0.014, effectively zero, and the scatter shows

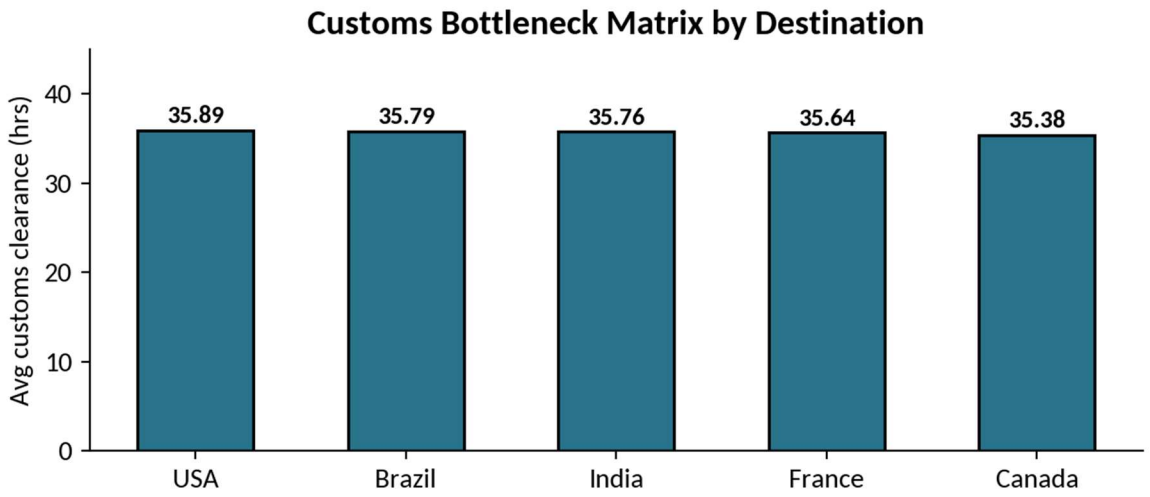
the same spread of 1 to 29 day lead times at every distance from 100 km to almost 15,000 km. Total distance shipped is 43.67 million km (Air 15.23M, Sea 14.56M, Land 13.88M), and the three modes cover similar average distances. In a real logistics network a long-haul order would normally take noticeably longer than a short one, so this flat pattern indicates that lead time is determined by factors other than distance. The dashboard plots one aggregated point per mode; the chart here shows every order to make the absence of a trend visible. Delay and lead-time targets should not be set purely by distance band until the underlying drivers are understood.

Delay Drivers Breakdown



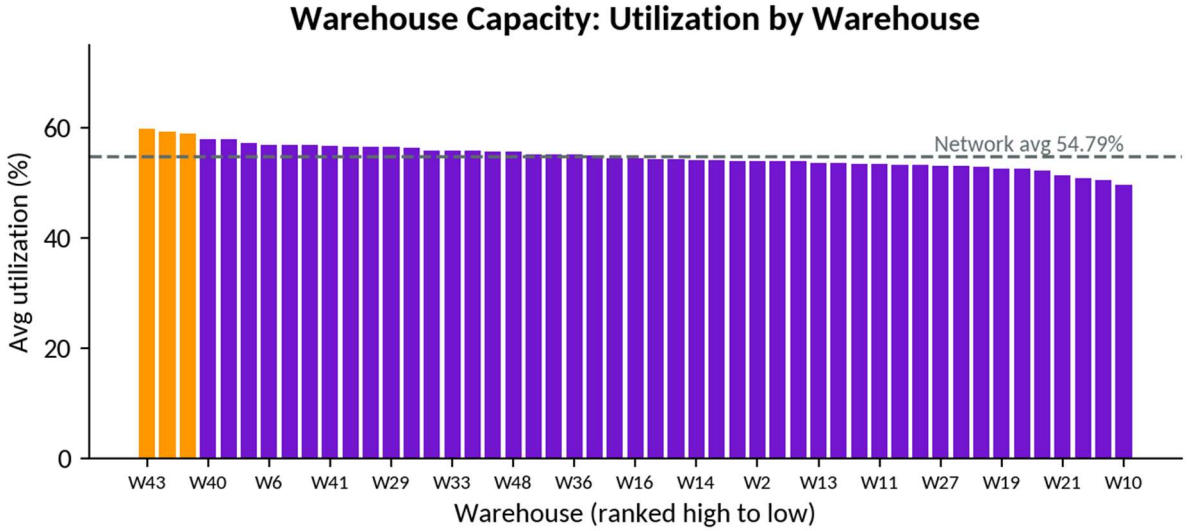
Delayed orders are spread across the whole day, with only mild peaks. The highest delay count falls in time slot 20 with 140 delayed orders (a 54.9% delay rate), followed by slot 19 with 134 (54.3%). The quietest slots are slot 4 with 101 delayed orders and slot 22 with 107; by rate, slot 9 is best at 42.3%. Within each slot the contribution of Air, Sea and Land fluctuates without a consistent leader. The month view is similarly balanced: delay rates range from 45.8% in April to 51.6% in July. Operationally, this suggests the late-evening slots (19–20) merit a modest staffing or dispatch review, but that delays are a general feature of the process rather than a symptom of one hour, mode or month.

Customs Bottleneck Matrix



Customs clearance takes an average of 35.69 hours per order, and it is nearly the same for every destination. USA is the slowest at 35.89 hours and Canada the fastest at 35.38 hours — a difference of only 0.51 hours — with Brazil (35.79), India (35.76) and France (35.64) in between. Individual orders, however, range from 1 to 71 hours, so the variation sits inside each destination rather than between them. Customs time also does not predict lateness: its correlation with the delay flag is 0.006. The matrix therefore does not identify a single problem country; the efficiency gain would come from reducing the long-tail clearances (orders above roughly 60 hours) across all destinations rather than from targeting one border.

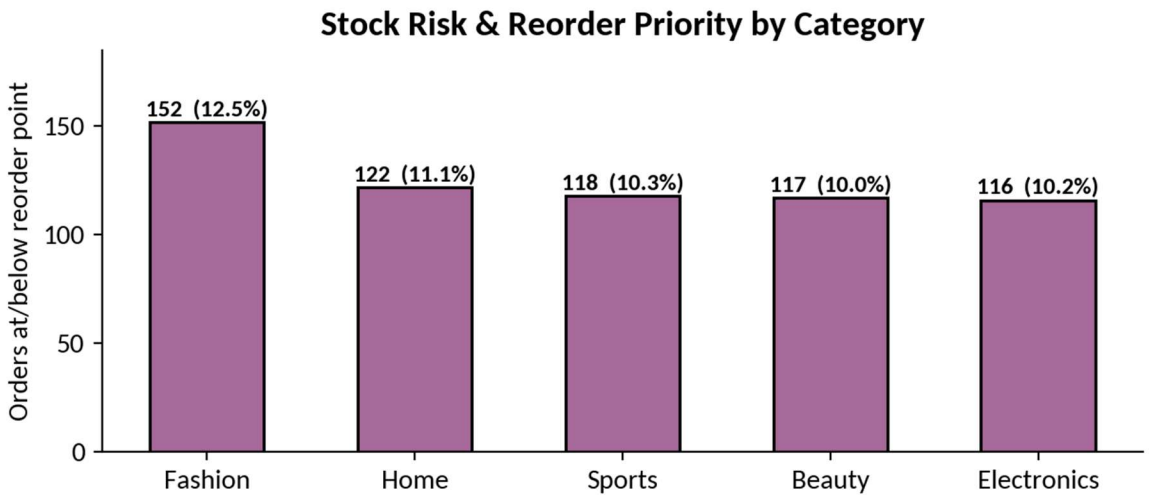
Warehouse Capacity



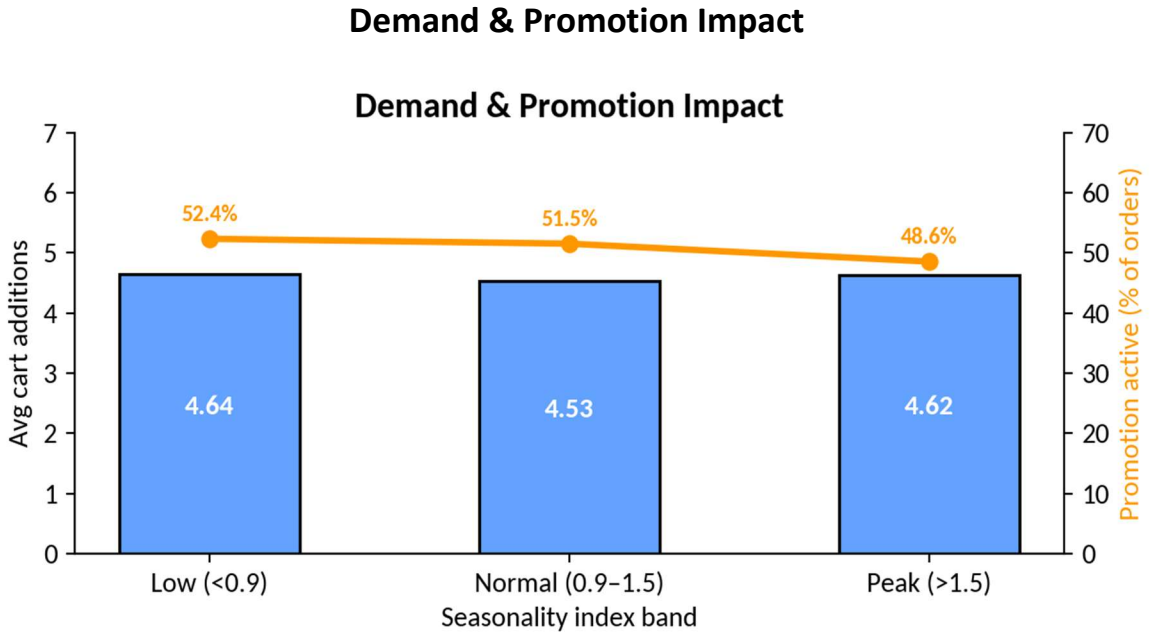
Warehouse utilization averages 54.79% across the network of 49 warehouses, leaving substantial headroom on average. Individual warehouses are tightly grouped between 49.6% (Warehouse 10) and

59.8% (Warehouse 43), so there is no clearly overloaded site and no obviously idle site; Warehouses 43, 49 and 38 are the three fullest at 59.8%, 59.4% and 59.0%. At the order level, however, 21.1% of orders were fulfilled from a warehouse reading above 80% utilization, which is the kind of peak load that average figures hide. The dashboard's pivot sums utilization per warehouse, which mainly reflects how many orders each warehouse handled; averages are used here so sites are comparable. Capacity planning should watch peak readings rather than the 54.79% network average.

Stock Risk & Reorder Priority by Category



Stock risk is modest and spread across categories. 625 order records (10.8%) show stock at or below the reorder point, and these involve 507 distinct products — the 507 items shown in the dashboard's Items Below Reorder Point card. Fashion carries the most at-risk records with 152 (12.5% of its orders), followed by Home (122), Sports (118), Beauty (117) and Electronics (116). One data-model note: the dashboard's Reorder_Count column holds the same value (507) on every row, so a bar chart that sums it simply mirrors order counts per category; the chart here counts the rows that are actually at or below the reorder point. Replenishment should be prioritized for Fashion first, with the other four categories needing similar attention.



Promotions and seasonality show very little relationship with customer engagement. Promotions are active on 2,928 orders (50.8%), and average cart additions are 4.61 with a promotion versus 4.58 without — a difference of only 0.03. Grouping the seasonality index into Low (below 0.9), Normal (0.9–1.5) and Peak (above 1.5) bands, cart additions stay between 4.53 and 4.64, and the correlation between seasonality and cart additions is -0.011. Peak-season orders are, if anything, slightly less likely to be promoted (48.6% versus 52.4% in the low band). One cross-variable point does stand out: promoted orders are delayed more often (50.5%) than non-promoted ones (47.6%), so promotional volume may add operational strain even though it does not measurably lift cart activity in this sample.

Summary of Key Metrics

Category	Metric	Value	% of Total
Overall KPIs	Total Orders	5,767	—
	Total Order Value (USD)	$5,732,623	—
	Average Order Value	$994.04	—
	Avg Delivery Lead Time	15.16 days	—
	Overall Delay Rate	49.09%	—
	On-Time Rate	50.91%	—
	Total Holding Cost	$59,240	—
	Total Distance Shipped	43,671,964 km	—
	Avg Customs Clearance	35.69 hrs	—
	Avg Warehouse Utilization	54.79%	—
	Items Below Reorder Point	507	—

Category	Metric	Value	% of Total
	Active Promotion Orders	2,928	50.8%
Product Category	Fashion (value)	$1,222,777	21.3%
	Beauty (value)	$1,161,000	20.3%
	Electronics (value)	$1,126,558	19.7%
	Sports (value)	$1,118,216	19.5%
	Home (value)	$1,104,073	19.3%
Shipping Mode	Air (orders)	1,953	33.9%
	Land (orders)	1,872	32.5%
	Sea (orders)	1,942	33.7%
Customer Segment	New (orders)	1,954	33.9%
	Premium (orders)	1,929	33.4%
	Regular (orders)	1,884	32.7%
Destination	France (orders)	1,186	20.6%
	India (orders)	1,165	20.2%
	USA (orders)	1,135	19.7%
	Canada (orders)	1,126	19.5%
	Brazil (orders)	1,155	20.0%
Month	January (orders / delay rate)	744 / 49.3%	12.9%
	February (orders / delay rate)	672 / 50.9%	11.7%
	March (orders / delay rate)	744 / 48.4%	12.9%
	April (orders / delay rate)	720 / 45.8%	12.5%
	May (orders / delay rate)	744 / 48.5%	12.9%
	June (orders / delay rate)	720 / 49.6%	12.5%
	July (orders / delay rate)	744 / 51.6%	12.9%
	August (orders / delay rate)	679 / 48.6%	11.8%

Next Steps to Move From Descriptive → Predictive

This report only describes what already happened in the dataset. If supply chain and logistics leaders wanted to act on these patterns going forward, the logical next layer of work would include:

- Delay Risk Modeling: Building classification models (using time slot, shipping mode, customs time, destination and promotion status) to score each order's probability of delay, going beyond today's 49.1% baseline.
- Lead Time Prediction: Regression models to estimate delivery lead time per lane, testing why distance and mode explain so little of it today.

- Stock-Out & Reorder Forecasting: Forecasting demand by category so replenishment starts before stock falls below the reorder point (currently 507 items).
- Promotion Impact Analysis: Uplift modeling to test whether promotions change purchasing behavior or simply add delivery pressure.

Any of these would build directly on the KPIs, category rankings and cross-tabs already produced here, once a live shipment and inventory event feed replaces the current order-level historical dataset.

Key Findings

- Scale: 5,767 orders worth $5.73M (average $994 per order) were shipped over eight months, with a 15.16-day average lead time, $59,240 in holding cost and 43.7 million km travelled.
- Delay is the central operational issue: 49.1% of orders are delayed (2,831 orders), and the rate is almost identical across modes (48.3%–49.5%), segments, destinations and months.
- Mode and distance do not drive speed: Air (15.03 days), Sea (15.10) and Land (15.37) are within a few hours of each other, and distance correlates -0.014 with lead time.
- Even business mix: product categories (19.3%–21.3% of value), customer segments (32.8%–33.7%) and destinations (19.6%–20.8%) are all close to equal shares, led by Fashion, New customers and France respectively.
- Customs and warehouses: customs clearance averages 35.69 hours with little difference by destination, and warehouse utilization averages 54.79% (49.6%–59.8% by site), though 21.1% of orders were handled at above 80% utilization.
- Supply chain strategy is the one field that does separate performance: Fast Fulfillment orders (762 orders) average 2.99 days versus 16.8–17.4 days for the other three strategies — yet its delay rate (50.4%) is no better, which suggests the delay flag is measured independently of lead time. This field is not shown in any dashboard visual.
- Dashboard text vs. data: the Executive header ticker states 'Zero Critical Bottlenecks Detected' and 'Revenue Growth: On Track', and the Logistics ticker shows On-Time Delivery Rate as 'Active'. These are fixed text strings, not calculated values; the data itself shows a 49.1% delay rate, and no month-over-month revenue growth measure exists in the model.
- Data-model notes: Reorder_Count (507) and Shipping Share % (5,767) hold one constant value on every row, so visuals that sum or count them reflect order counts rather than true reorder or share figures; converting them to measures would fix this.

References

- Cross-Border Logistics & Supply Chain Control Dashboard (Executive, Logistics, Inventory and Dataset pages), built in Power BI — source file n.pbix.

- cross_border_supply_chain_dataset (n = 5,767 orders, 43 fields), extracted from the Power BI data model.
- Course materials: Business Analytics — Descriptive Analytics Assignment Brief.

Appendix — Dashboard & Raw Data

The Power BI file is a four-page report. The table below lists each page and the visuals it contains, so that every chart in this report can be traced to the page it comes from. Each KPI card and chart in this report was computed from the record-level dataset described in the sample that follows.

Page	Visuals
Executive	Total Order Value, Avg Delivery Lead Time, Overall Delay Rate and Total Holding Cost cards; Revenue & Performance by Product Category; Transport Performance & Bottleneck Analysis; Customer Segment Breakdown; Global Shipping Route Map; slicers for Customer Segment, Product Category, Shipping Mode and Month
Logistics	Avg Customs Time, Air/Sea/Land Share and Total Distance Shipped cards; Distance vs. Lead Time; On-Time Delivery Rate by Shipping Mode; Delay Drivers Breakdown; Customs Bottleneck Matrix; four slicers
Inventory	Total Warehouse, Avg Warehouse Utilization, Items Below Reorder Point and Total Active Promotions cards; Warehouse Capacity pivot; Demand & Promotion Impact; Stock Risk & Reorder Priority by Category; four slicers
Dataset	Full order-level table with 33 fields and four slicers (Customer Segment, Product Category, Shipping Mode, Month)

Sample of Underlying Record-Level Data

While the dashboard shows only pre-aggregated summaries, every KPI card and chart in this report was computed from the underlying cross_border_supply_chain_dataset — one row per order, with fields for product, customer segment, origin and destination country, shipping mode, distance, customs clearance time, lead time, delay status, warehouse, stock, promotion and demand. The table below shows the first 30 of the 5,767 rows (sorted by Order ID) so that the calculation behind any KPI can be spot-checked by hand. Two reconciliation checks tie this data to the dashboard: the sum of Order_Value across all rows is $5,732,622.51, and the sum of Delivery_Lead_Time divided by the row count gives the 15.16-day average shown on the Executive page.

Order ID	Date	Category	Value ($)	Segment	Route	Mode	Km	Cust. hrs	Lead days	Delay
100000	01-Jan-23	Sports	1,203.68	New	Germany → Brazil	Air	11,887	65	4	0
100001	01-Jan-23	Home	324.79	Regular	China → Brazil	Land	495	14	25	0
100002	01-Jan-23	Electronics	425.15	Regular	Germany → India	Air	5,422	35	13	0
100003	01-Jan-23	Electronics	569.36	New	UK → India	Air	4,415	52	20	1
100004	01-Jan-23	Electronics	1,021.48	Premium	Germany → India	Sea	4,614	23	17	0
100005	01-Jan-23	Fashion	1,024.06	Regular	China → Canada	Air	3,397	20	26	1
100006	01-Jan-23	Electronics	1,629.52	New	Germany → USA	Air	5,433	27	13	0
100007	01-Jan-23	Fashion	1,507.26	Premium	UK → Canada	Air	2,506	41	22	1
100008	01-Jan-23	Electronics	1,827.90	Regular	USA → Canada	Air	8,792	36	16	0
100009	01-Jan-23	Home	1,666.64	Regular	UK → USA	Sea	3,642	4	13	1
100010	01-Jan-23	Sports	511.49	Regular	Germany → France	Land	8,047	26	23	1
100011	01-Jan-23	Sports	665.94	New	India → France	Land	13,430	40	1	0

Order ID	Date	Category	Value ($)	Segment	Route	Mode	Km	Cust. hrs	Lead days	Delay
100012	01-Jan-23	Electronics	1,003.34	Regular	Germany → France	Air	11,714	66	25	0
100013	01-Jan-23	Sports	33.47	Regular	Germany → India	Land	12,248	3	10	1
100014	01-Jan-23	Electronics	1,808.28	Premium	China → India	Land	11,120	64	18	0
100015	01-Jan-23	Electronics	93.47	New	India → Brazil	Air	9,078	29	23	0
100016	01-Jan-23	Beauty	648.27	Premium	China → Brazil	Air	13,063	54	27	1
100017	01-Jan-23	Beauty	1,140.50	Regular	China → Canada	Land	2,361	7	24	1
100018	01-Jan-23	Beauty	1,038.09	New	UK → Brazil	Sea	2,639	25	24	0
100019	01-Jan-23	Fashion	1,310.65	New	USA → India	Land	8,513	24	17	1
100020	01-Jan-23	Beauty	1,057.40	Regular	Germany → Canada	Air	2,023	43	26	1
100021	01-Jan-23	Electronics	461.86	New	India → Brazil	Land	6,512	6	17	0
100022	01-Jan-23	Home	406.15	New	Germany → Brazil	Land	1,207	15	10	0
100023	01-Jan-23	Beauty	1,814.96	Premium	China → France	Sea	4,938	40	20	0
100024	02-Jan-23	Beauty	531.61	New	USA → India	Land	6,121	61	2	0
100025	02-Jan-23	Home	565.55	New	China → Canada	Land	2,191	21	10	0
100026	02-Jan-23	Electronics	828.84	Premium	Germany → USA	Air	1,711	58	28	0
100027	02-Jan-23	Fashion	283.18	Regular	UK → Brazil	Sea	5,171	32	20	0
100028	02-Jan-23	Beauty	1,017.01	Premium	Germany → India	Land	4,048	12	6	0
100029	02-Jan-23	Home	1,331.74	Regular	China → Canada	Land	8,116	9	3	0